
# B.PASINDU MANODARA BOLONGHE

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## Profile Summary

I am a forward-thinking person driven by a passion for innovation and a deep curiosity about where technology meets creativity. I focus on virtual reality (VR) application development, using it to create immersive experiences that solve real-world problems, while also exploring machine learning and cybersecurity. I welcome challenges as chances to learn and grow.

VR Application Development | Machine Learning | Cybersecurity

## Projects & Researches

**Demo: Compiling Adaptive VR Simulations from Conceptual Content in Real Time** *2026*
This is a demo paper introduces a system that generates VR simulations from conceptual inputs, including text, diagrams, and equations. The system extracts variables, relationships, and constraints from input content and compiles them into executable Unity environments without requiring manual specification of simulation logic. Rather than producing a single static simulation, the system generates simulations at progressively increasing levels of conceptual complexity derived from the same input, enabling "what-if" exploration across levels. The paper was published and presented in **MobiSys 2026**.

**Toward Stateful Generative Environments from Structured Content** *2026*
This is a vision paper introducing how to implement virtual reality simulations from structured content without manual authoring. The suggested pipeline consists of 5 LLM agents which are working together to create interactive VR simulation. The environment monitors user behavior through eye tracking, head movement and interaction monitoring to provide feedback to the learner. This paper was accepted and presented in **AutoSys 2026** workshop which was co-located with **MobiSys 2026**.

**Motor-Mediated Creativity: Bridging Embodied Skill Training and Digital Expression** *2025*
This research introduces motor-mediated creativity, a paradigm that cultivates motor fluency to enhance expressive digital drawing by addressing sensorimotor bottlenecks in novices. Through Motus, an interactive system, we integrated explicit motor training (e.g., velocity- and pressure-modulated strokes) with ideation prompts to reduce cognitive load in translating intent to output, grounded in embodied cognition and motor learning. Formative and evaluation studies demonstrated improved expressive agency and artwork quality, submitted to **CHI2026**.

HCI | Motor Learning | Expressive Motor Control

**Competitive virtual reality cycling over low latency networks** *2025*
As team leader for our final-year project, I directed the creation of an innovative VR cycling simulator emphasizing seamless navigation and realistic motion dynamics. The system delivers competitive multiplayer racing with low-latency networking for synchronized rider positioning, immersive 360-degree environmental traversal, and photorealistic real-world terrains. Key to our motion fidelity, we integrated adaptive resistance feedback in the initial prototype, simulating dynamic uphill pedaling and downhill momentum to enhance proprioceptive realism. Hosted at the Dialog Innovation Center in the Lotus Tower for imminent public demonstration, my contributions centered on leading the Oculus VR application in Unity 3D, optimizing gesture-based steering for intuitive path navigation and fluid body-motion tracking. Our associated research paper on Virtual Reality Cycling: a Competitive and Low-Latency Exer-Gaming Environment was presented at IEEE Gaming , media & Entertainment (**IEEE GEM2025**) conference 2025.

Unity 3D | c# | Motor Navigation | IoT

**Machine learning project** *2024*
As a part of my personal interests, I engage in machine learning projects focusing on image and text classifications and model tuning . For a detailed showcase of my work, you can explore my GitHub account.

tensorflow | numpy | pytorch

# Working Experience

**Singapore Management University** *Jan 2026 - Now*
**Research Engineer**
Implement agentic systems to generate VR simulations from text book contents (eg: images, paragraphs or equations). Research about pipeline accuracy and user interactions with generated simulations.
C# | Unity | LLM

**Thakshana Technologies (Pvt) Ltd** *Jun 2024 - Dec 2025*
**Embedded software Engineer**
Design & develop APIs, communication protocols, and cryptographic applications for embedded systems
C++ | c | Certificates | Signatures

**KATIM Abu Dhabi,UAE** *oct 2024, Nov 2025*
**Embedded software Engineer**
Got a training in KATIM in embedded software engineering

**4Axis Solutions (Pvt) Ltd** *Jan 2023 - June 2023*
**Software Engineering Intern**
I was a member of the VR team where I actively contributed to the development of a drawing application named Drawing Desk specifically designed for Oculus VR devices. During my internship, I acquired proficiency in Unity 3D and C#, skills that I continue to apply and enhance through my ongoing commitment to learning, particularly in the context of my final year project. The application which we developed can be found here.
Unity 3D | c# | Unity shaders | Database - firebase

# Education

**University of Moratuwa** Moratuwa, Sri Lanka
*B.Sc. Engineering (Hons.) in Electronics and Telecommunications* *Dec 2019 - Dec 2024*
*CGPA: 3.70 (First class)*

**D.S Senanayake College** Colombo 07, Sri Lanka
*GCE Advanced Level - Physical Science Stream* 2016-2018
*Z-Score 2.1817*

**Prince Of Wales college** Moratuwa, Sri Lanka

# Skills Summary

**Languages**: Python, C++, C#, C, matlab, Arduino
**Tools**: Unity 3D, Solidworks, Altium
**Soft Skills**: Leadership, Organizing, Event Management
**Technical Skills**: OpenCV, tensorflow, numpy, VS code, Git

# Achievements and Awards

**SPARK Challenge 21/22**: Top 10 finalists
**IEEE Signal Processing Cup 2022**: Worldwide 21st Place
**Future Innovators Challenge'22** : Semi finalist

# Courses and certificates

**DeepLearning.AI TensorFlow Developer Professional Certificate** *2022*
Introduction to tensorFlow for artificial intelligence, machine Learning, and deep Learning

**DeepLearning.AI Deep Learning Specialization** *2023*
Improving deep neural networks: hyperparameter tuning, regularization and optimization

**Data science foundations: Data structures and algorithms specialization** *2023*
University of Colorado Boulder

**HCIA-AI course** *2021*
Huawei AI course(HCIA-AI) by Huawei talent Online in University of Moratuwa ICT Academy

## Leadership and Volunteering

**Rotaract Club - University of Moratuwa** *2020-2021*
Co-chairperson of "Andurata Athwelak" computer course project launched by Rotaract Mora. That project was awarded the most outstanding Rotaract project of the year 2020/2021

**Electronic Club - University of Moratuwa** *2022-2023*
Social media coordinator

## References

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